

A Hybrid Adaptive-Network Model of Algorithmically Mediated Opinion Dynamics

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Abstract

With online polarization becoming an increasing concern, understanding the underlying opinion dynamics is essential for identifying the mechanisms that produce and reinforce collective division. However, modeling these processes comes with many difficulties due to the interacting effects of psychology, complex social networks, and algorithmic mediation. This study develops a hybrid agent-based model of online opinion dynamics that combines algorithmic content recommendation, social influence, bounded confidence, dynamic stubbornness, and adaptive network rewiring. The model was evaluated through mechanism checks, six interpretable social-media environments, and a randomized search across plausible parameter combinations. Our findings suggest that polarization emerges from interactions among algorithmic selectivity, user receptiveness, and evolving social structure rather than from any single mechanism in isolation.

1 Introduction

Opinion dynamics studies how individual opinions evolve over time, and how repeated local or global interactions produce global outcomes (consensus, fragmentation, polarization). In opinion dynamics models, a population is represented as a set of agents, each holding an opinion that changes according to an update rule. Tomé, Fiore, and de Oliveira describe opinion dynamics as the time evolution of the number of individuals in different opinion groups, where opinions change due to the influence of other individuals or external agents (Tomé et al., 2023).

In a graph-based opinion dynamics model, an individual’s opinion at a given time can be represented as

$$x_i(t) \in \mathcal{O}, \quad (1)$$

where $x_i(t)$ is agent i ’s opinion at time t , and $\mathcal{O}$ is the opinion space. The evolution of an agent’s opinion can then be expressed generally through an update rule,

$$x_i(t+1) = \Phi(G(t), \mathbf{x}(t), i), \quad (2)$$

where Φ determines the next opinion from the current social graph, the population’s current opinions, and the characteristics of agent i .

Many classic opinion-dynamics models isolate social influence, local network interactions, or bounded-confidence updating, rather than representing all of these mechanisms simultaneously (Shirzadi, n.d.). After initially exploring a variety of discrete and continuous, graph and non-graph based

models, this study focuses on a hybrid extension that combines continuous 1-dimensional opinions, user-user influence, user-content exposure, and adaptive network rewiring in a singular model that has additional parameters to simulate heterogeneous user behavior, algorithmic exposure, worldview consolidation, and the co-evolution of opinions and social structure. Simulations of the model suggest that polarization is influenced not by one parameter alone, but by interactions among user openness, algorithmic personalization, social influence, and rewiring rate of social networks

2 Model formulation

Our model is a hybrid agent-based model that simulates shifts in opinion that lie on a continuous domain, and combines algorithmic content recommendation with social influence over an adaptive user network. $x_i(t) \in [-1, 1]$, and M content items with fixed positions $y_j \in [-1, 1]$. The user-user social structure is represented by a graph $G(t)$, while content items are stored as a separate array rather than as persistent graph nodes. Social edges therefore represent lasting but also evolving relationships, whereas user-content interactions are temporary exposure events generated at each time step.

The evolving state is represented as

$$\mathcal{X}(t) = (\mathbf{x}(t), G(t), \mathbf{z}(t), \boldsymbol{\tau}(t), \mathbf{h}(t)), \quad (3)$$

where $\mathbf{x}(t)$ contains user opinions, $G(t)$ is the social graph, $z_i(t)$ is user i 's current worldview center, $\tau_i(t)$ is the time spent near that worldview, and $h_i(t)$ is the user's current stubbornness.

2.1 Algorithmic content exposure

The recommender assigns an exposure probability to each user-content pair:

$$P_{ij}(t) = (1 - r) \frac{\exp[-\beta_i |x_i(t) - y_j| + q \log(p_j + c)]}{\sum_{m=1}^M \exp[-\beta_i |x_i(t) - y_m| + q \log(p_m + c)]} + \frac{r}{M}, \quad (4)$$

where β_i controls personalization, y_j is the opinion position of content item j , p_j is its popularity, q controls popularity strength, $c > 0$ prevents the logarithm of zero, and $r \in [0, 1]$ is the exploration rate. The first component is a softmax distribution over a score combining ideological similarity and popularity, and the second component introduces uniform random exploration.

At each time step, K items are sampled from this distribution. Let $\mathcal{R}_i(t)$ denote the set shown to user i , and define

$$w_{ij}(t) = \frac{P_{ij}(t)}{\sum_{m \in \mathcal{R}_i(t)} P_{im}(t)}. \quad (5)$$

Following bounded-confidence dynamics, only content within user i 's confidence threshold ε_i affects their opinion. The content pull is

$$C_i(t) = \sum_{j \in \mathcal{R}_i(t)} w_{ij}(t) \mathbf{1}\{|y_j - x_i(t)| \leq \varepsilon_i\} [y_j - x_i(t)]. \quad (6)$$

2.2 Social influence

Define the set of persuasive social neighbors by

$$\mathcal{N}_i^\varepsilon(t) = \{k \in \mathcal{N}_i(G(t)) : |x_k(t) - x_i(t)| \leq \varepsilon_i\}. \quad (7)$$

The resulting social pull is

$$S_i(t) = \begin{cases} \frac{1}{|\mathcal{N}_i^\varepsilon(t)|} \sum_{k \in \mathcal{N}_i^\varepsilon(t)} [x_k(t) - x_i(t)], & |\mathcal{N}_i^\varepsilon(t)| > 0, \\ 0, & |\mathcal{N}_i^\varepsilon(t)| = 0. \end{cases} \quad (8)$$

2.3 Opinion update

Each user has an activity probability a_i , with

$$u_i(t) \sim \text{Bernoulli}(a_i). \quad (9)$$

The complete opinion update is

$$x_i(t+1) = \Pi_{[-1,1]} \left[x_i(t) + u_i(t) \left(\mu_c s_i C_i(t) + \mu_s s_i S_i(t) + A_i(t) + H_i(t) + N_i(t) \right) \right], \quad (10)$$

where $\Pi_{[-1,1]}$ projects the result onto the permitted opinion range, s_i is user-specific susceptibility, and μ_c and μ_s control the overall strengths of content and social influence. The remaining terms are defined as

$$A_i(t) = \alpha e_i x_i(t) [1 - |x_i(t)|], \quad (11)$$

$$H_i(t) = \lambda h_i(t) [z_i(t) - x_i(t)], \quad (12)$$

$$N_i(t) \sim \mathcal{N}(0, \sigma^2), \quad (13)$$

where e_i is extremity bias, α controls amplification, λ controls anchoring toward the worldview center, and $N_i(t)$ represents random external influence.

2.4 Dynamic stubbornness

When

$$|x_i(t+1) - z_i(t)| \leq \delta, \quad (14)$$

the user's residence time and worldview center update according to

$$\tau_i(t+1) = \tau_i(t) + 1, \quad (15)$$

$$z_i(t+1) = (1 - \eta) z_i(t) + \eta x_i(t+1). \quad (16)$$

A movement outside the stability region resets the residence time, $\tau_i(t+1) = 0$, and establishes a new worldview center by resetting $z_i(t+1) = x_i(t+1)$. Current stubbornness is

$$h_i(t+1) = \Pi_{[0, h_{\max}]} \left[h_i^{(0)} + \gamma \left(1 - \exp \left[-\frac{\tau_i(t+1)}{T_c} \right] \right) \right], \quad (17)$$

where γ controls consolidation strength and T_c controls its timescale. This represents worldview consolidation and the increasing resistance to change that may develop after prolonged commitment to a worldview. It is motivated by psychological processes such as confirmation bias, belief perseverance, and identity-protective reasoning.

2.5 Dynamic homophily

At fixed rewiring intervals, a fraction of non-bridge edges is replaced. Existing ties are selected for removal according to

$$P_{\text{remove}}(i, k) \propto \exp(\kappa |x_i(t) - x_k(t)|) - 1, \quad (18)$$

while new ties are sampled according to

$$P_{\text{add}}(i, k) \propto \exp(-\kappa |x_i(t) - x_k(t)|). \quad (19)$$

The homophily parameter κ controls the preference for similar users. The rewiring fraction controls the magnitude of network change, and the rewiring interval controls its frequency. Bridge edges are protected to preserve network connectivity, and each removed edge is replaced so that the overall network density remains approximately constant.

Together, these rules define a coupled stochastic discrete-time dynamical system in which opinions affect recommendations and social structure, and these mechanisms subsequently affect future opinions.

3 Experimental Methods

In each simulation, an initial population distribution was selected as either being uniform, bimodal, or normal, and specific values for the parameters were selected per simulation. The graph type is also selected as either being a small-world graph, a homophily graph, or a dynamic homophily graph. Different distributions were used for the different heterogeneous model parameters, with Beta distributions being used for asymmetric populations. Gamma and Pareto distributions were used for personalization and popularity, as both are nonnegative and right-skewed with a very small number of users/content receiving unusually large values.

Each simulation was run for T discrete timesteps, where at each time step, users became active according to their individual activity probabilities(asynchronous model). Active users receive K content items sampled from the recommender distribution(for the recommender equation of the selected parameters); content and social pulls are calculated according to each user's confidence threshold, and then opinions are updated according to the update equation.

To isolate the contributions and effects of the two influence mechanisms (content and social), we evaluated the model under three conditions: a content-only model with $\mu_c > 0$ and $\mu_s = 0$, a social-only model with $\mu_c = 0$ and $\mu_s > 0$, and a hybrid model with both $\mu_c > 0$ and $\mu_s > 0$.

Table 1: Initialization of heterogeneous model parameters.

Parameter	Distribution	Interpretation
ε_i	Beta	Confidence threshold
s_i	Beta	Susceptibility
β_i	Gamma	Personalization strength
a_i	Uniform	Activity probability
e_i	Beta	Extremity bias
p_j	Pareto	Content popularity
$N_i(t)$	Normal	Random external influence

While we did perform parameter sweeps for all parameters, we first evaluated the model under six interpretable parameter regimes to compare plausible social-media environments in which multiple algorithmic, psychological, and social mechanisms operate simultaneously. The regimes represented weak personalization, moderate personalization, algorithmic filtering, popularity-dominated exposure, socially dominated echo chambers, and cross-cutting high-openness environments.

Table 2: Qualitative characteristics of the simulated environments.

Regime	Personalization	Openness	Dominant influence
Diverse	Low	High	Mixed
Moderate	Medium	Medium	Mixed
Filter bubble	High	Low	Content
Mainstream	Low-medium	Medium	Popularity
Echo chamber	Medium	Low	Social
Cross-cutting	Low	High	Exploratory

The model’s results were evaluated using polarization, opinion switching, and network segregation measures. Polarization looked at the degree of separation in the final opinion distribution. Let T denote the final timestep and let N be the number of users. Polarization was measured by combining mean pairwise opinion distance with the mean distance from the ideological center. The mean pairwise distance at the final timestep, which measures how far apart users are from one another on average, is

$$D(T) = \frac{2}{N(N-1)} \sum_{i < j} |x_i(T) - x_j(T)|, \quad (20)$$

and final extremity, which measures how far the average user is from the "center", is

$$E(T) = \frac{1}{N} \sum_{i=1}^N |x_i(T)|. \quad (21)$$

The polarization score is then

$$\mathcal{P}(T) = D(T)E(T). \quad (22)$$

This measure is high only when users are both separated from one another and located far from the neutral center.

Ever-flip fraction counts how much users crossed from their original side to the opposite side at any point in the simulation, (values within a tolerance $\nu = 0.02$ of zero were treated as neutral). Define

$$\text{sgn}_\nu(x) = \begin{cases} +1, & x > \nu, \\ 0, & |x| \leq \nu, \\ -1, & x < -\nu. \end{cases} \quad (23)$$

Let

$$V_0 = \{i : \text{sgn}_\nu(x_i(0)) \neq 0\}$$

be the set of users whose initial opinions were non-neutral. The fraction that crossed to the opposite side at least once is

$$F_{\text{ever}} = \frac{1}{|V_0|} \sum_{i \in V_0} \mathbf{1} \{ \exists t \in \{1, \dots, T\} : \text{sgn}_\nu(x_i(t)) = -\text{sgn}_\nu(x_i(0)) \}. \quad (24)$$

Persistent switching measures the fraction that remained on the opposite side of their original side at the final timestep:

$$F_{\text{persistent}} = \frac{1}{|V_0|} \sum_{i \in V_0} \mathbf{1} \{ \text{sgn}_\nu(x_i(T)) = -\text{sgn}_\nu(x_i(0)) \}. \quad (25)$$

Network segregation was evaluated using the final social graph $G(T) = (V, E_T)$. Let

$$E_T^* = \{(i, j) \in E_T : \text{sgn}_\nu(x_i(T)) \neq 0 \text{ and } \text{sgn}_\nu(x_j(T)) \neq 0\}$$

denote edges whose endpoints both have non-neutral opinions. The same-side edge fraction is

$$S_{\text{edge}} = \frac{1}{|E_T^*|} \sum_{(i, j) \in E_T^*} \mathbf{1} \{ \text{sgn}_\nu(x_i(T)) = \text{sgn}_\nu(x_j(T)) \}. \quad (26)$$

We consider two connected users for each social edge (i, j) that are on the same side (neutral users are excluded). Values near one indicate that social ties occur primarily between users on the same side of the opinion space, while lower values indicate more cross-cutting connections. Outcomes were averaged across repeated simulation runs.

4 Results and Discussion

4.1 Preliminary Mechanism and Robustness Checks

Preliminary experiments isolated the content and social influence components of the model. Under the baseline parameterization, the content-only condition produced the greatest opinion variance(1) and disagreement(2), while the social-only and hybrid conditions produced slightly weaker changes. The simulations were also very sensitive to the initial opinion distribution, which has been suggested in previous papers (Turner and Smaldino, 2018). Initially bimodal populations retained substantially greater polarization(3) and extremity(4) than uniformly distributed populations. Static small-world, static homophily, and dynamically rewired networks produced similar aggregate opinion trajectories under otherwise identical parameters, suggesting that graph structure alone had less influence than the behavioral and algorithmic parameters in this specific setting. We also ran the hybrid model across different recommender parameters. Results indicated that stronger personalization tended to increase final opinion variance and disagreement, where higher exploration tended to decrease polarization.

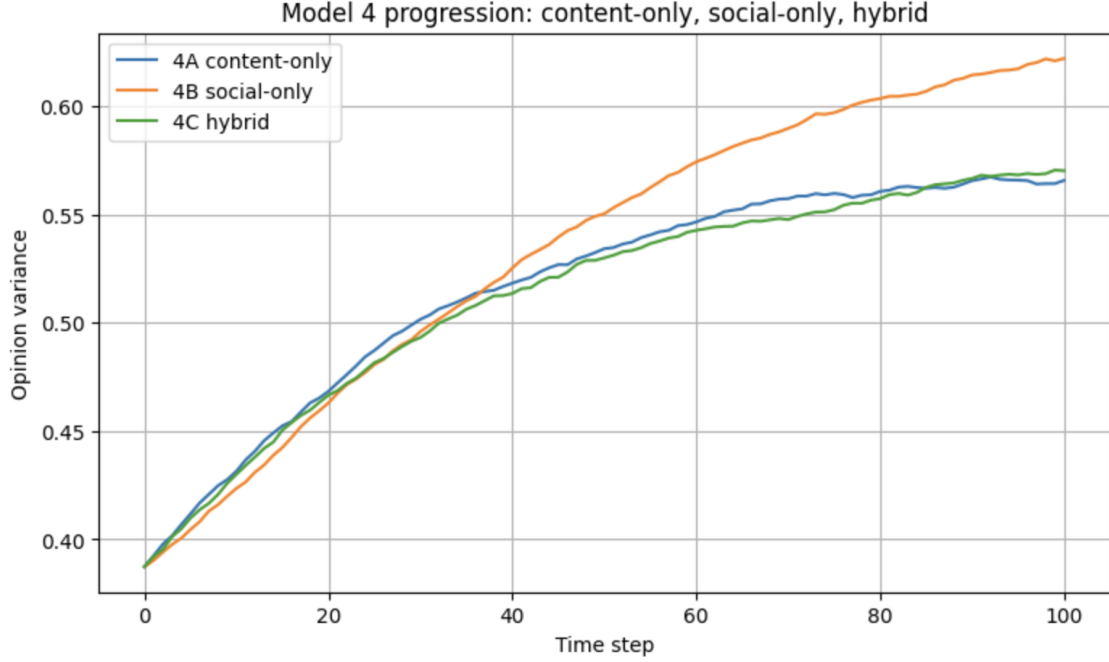


Figure 1: Comparison of opinion variance for models 4A, 4B, and 4C

4.2 Outcomes Across Interpretable Environments

We ran simulations for the hybrid model across six different interpretable parameter combinations.

The results showed that the socially dominated echo chamber and algorithmic filter-bubble environments produced the highest polarization scores, 0.686 and 0.677, respectively. The social graphs for these parameter combinations were also completely segregated and had no instances of opinion switching. While these two regimes emphasized different influence mechanisms, their strikingly similar outcomes indicate that algorithmic filtering and socially reinforced network segregation can provide alternative pathways to high polarization. The lowest polarization score of 0.248 came from the cross-cutting high-openness condition, and had the highest flip ratios as well.

Table 3: Primary outcomes across interpretable simulation regimes.

Regime	Polarization	Ever flip	Persistent flip	Same-side edges
Social echo chamber	0.686	0.000	0.000	1.000
Algorithmic filter bubble	0.677	0.000	0.000	0.999
Moderate personalization	0.597	0.003	0.003	0.739
Diverse weak personalization	0.472	0.042	0.034	0.631
Popularity-dominated mainstream	0.441	0.014	0.014	0.651
Cross-cutting high openness	0.248	0.376	0.141	0.604

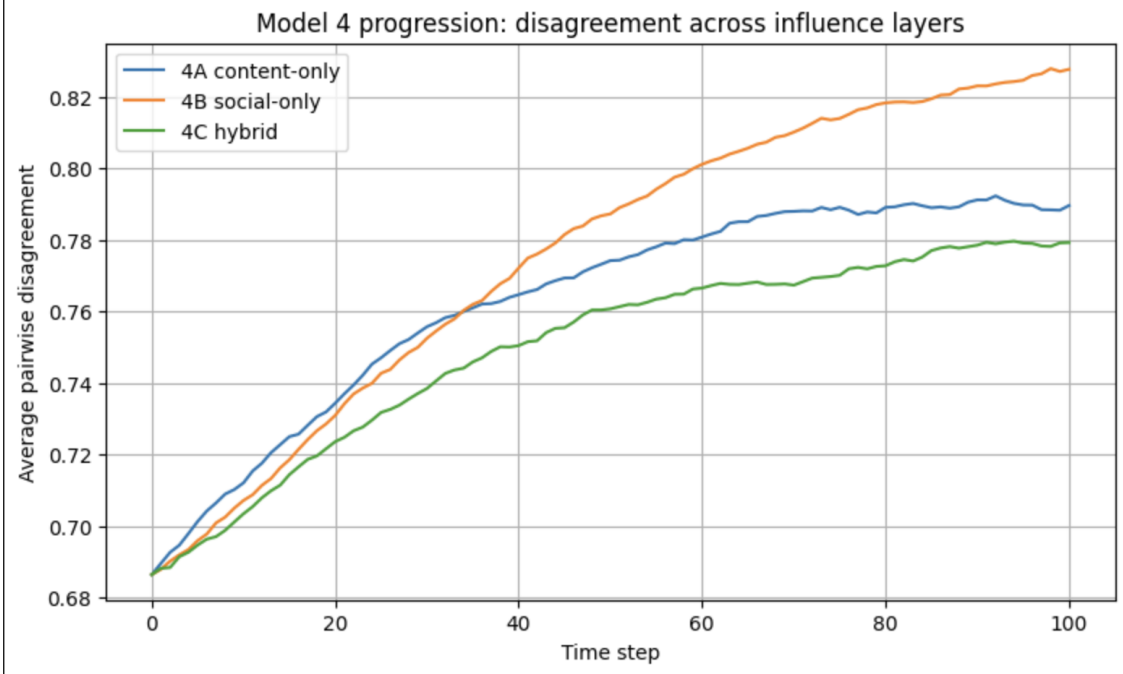


Figure 2: Comparison of pairwise disagreement for models 4A, 4B, and 4C

4.3 Associations Across the Randomized Parameter Search

The interpretable parameter regimes are effective in that they demonstrate activity in realistic system-level scenarios. However, they vary many parameters simultaneously, making it hard to distinguish what parameter changes lead to what changes. The results of the parameter sweeps gave the following results.

Table 4: Correlations between model parameters and simulation outcomes

Parameter	Polarization Mean	Ever Flip Mean	Persistent Flip Mean	Same-Side Edge Mean
Mean personalization, $\bar{\beta}$	0.4266	-0.3012	-0.2801	-0.1656
Popularity strength	-0.1078	0.0175	0.0316	0.1012
Exploration	-0.0339	0.0108	0.0154	0.0742
Content influence, μ_c	-0.0636	-0.1947	-0.2089	-0.1771
Social influence, μ_s	-0.2310	0.2584	0.2723	0.2027
Mean confidence threshold, $\bar{\epsilon}$	-0.7451	0.7261	0.7168	0.0716
Mean susceptibility	0.1800	-0.1807	-0.1816	0.2032
Homophily strength	-0.0707	0.2088	0.2066	0.0808
Rewiring fraction	0.2964	-0.2960	-0.3016	0.5016
Rewiring interval	0.0427	0.0488	0.0754	-0.5013
Consolidation strength	0.1691	0.0504	0.0555	0.0211

The parameter that showed the strongest relationship between polarization and opinion switching metrics was the confidence threshold, which represents user openness. Broader thresholds led to lower polarization and more frequent opinion switching. These results suggest receptiveness/openness to dissimilar information was more consequential than just exposure alone.

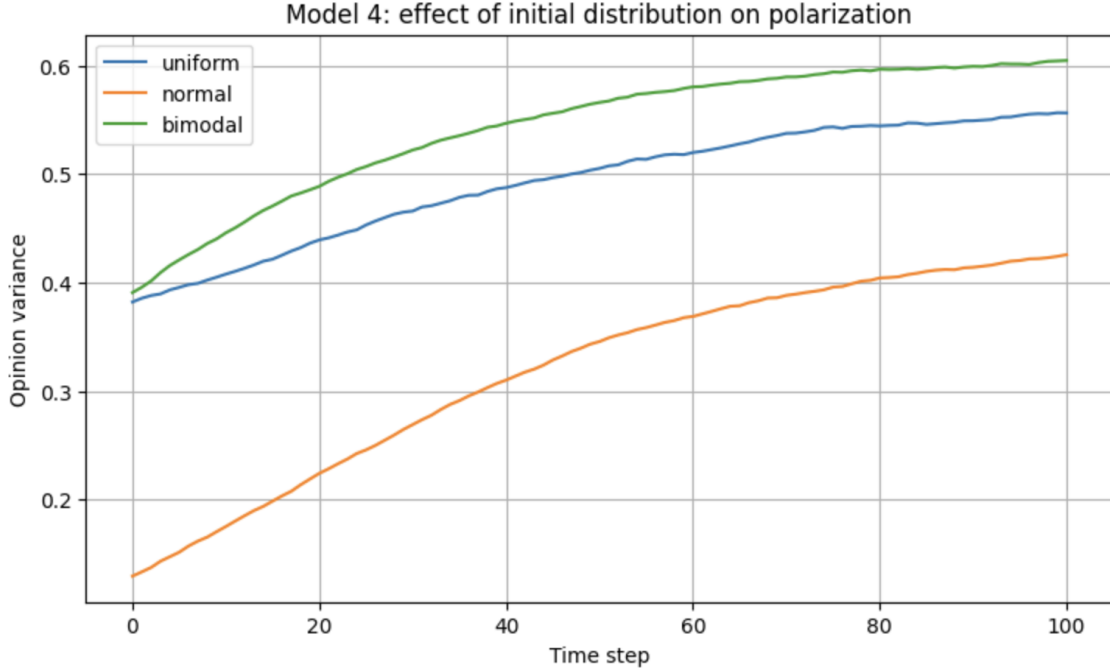


Figure 3: Polarization across different initial distributions

While not as strong, strong personalization was associated moderately with greater polarization and lower rates of side-switching. This follows, since repeated exposure to ideologically similar content would make people more resistant to changing their worldviews (Epstein et al., 2025).

Rewiring parameter values also had moderate correlations, with there being a moderate positive association between rewiring fraction and polarization, and a moderate negative association between rewiring interval and social segregation.

Together, the results suggest that polarization emerges when selective exposure is combined with narrow confidence thresholds and sufficient network adaptation to repeatedly reinforce existing divisions. Conversely, diverse exposure produces substantial opinion change primarily when users remain receptive to positions outside their existing worldview. The results also suggest that algorithmic filter bubbles and socially reinforced echo chambers can provide different pathways to similar collective outcomes of strong segregation between opinion groups.

5 Limitations

It is important to consider that opinions were represented on a highly simplified one-dimensional axis, users and content were generated from assumed distributions, and we used a simplified recommender interpretable model instead of complex, data-driven ranking systems commonly used by modern social-media platforms. With all of these simplifications, these findings should be interpreted as qualitative, not predictive. With no calibration with real-world data, it is hard to ground these parameter associations in real numbers. In addition, the associations mentioned in this paper are purely marginal, and do not capture nonlinear relationships, which would require further analysis.

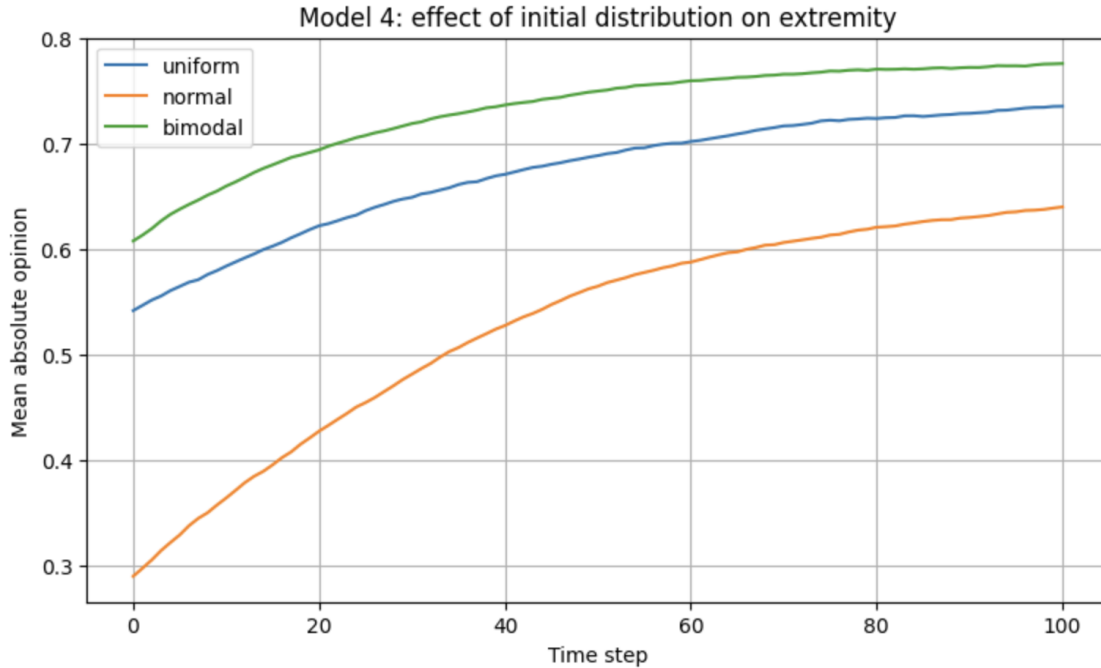


Figure 4: Extremity across different initial distributions

6 Final conclusions

Despite these limitations, our results demonstrate the value of modeling online opinion dynamics as a coupled hybrid human-algorithm dynamical system. Recommendation algorithms determine what content users are exposed to, and the evolving social network determines which interpersonal influences are repeatedly available and reinforced over time. Our results show that high polarization and low opinion switching tend to be correlated with strong algorithmic personalization, narrow confidence thresholds, and more extensive network rewiring, but uncovering the more implicit, and perhaps more revealing relationships between these parameters will require further analysis.

References

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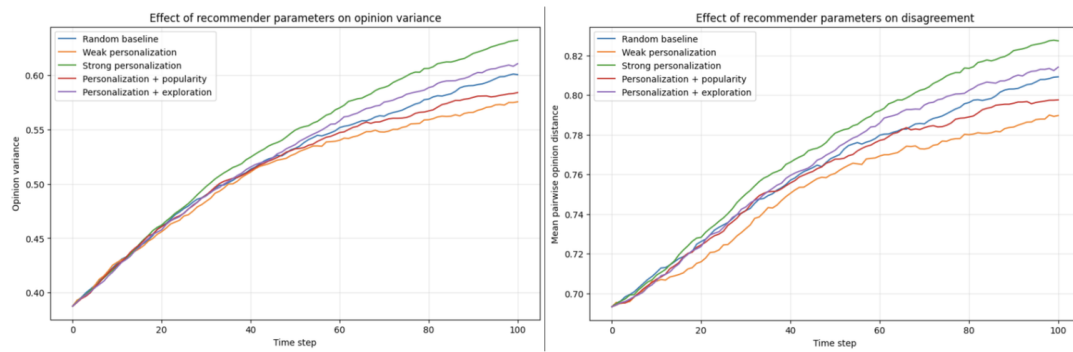


Figure 5: Effect of recommender parameters on opinion variance and disagreement.